

Acoustic Emission and Passive Acoustic Regression to the ISO 281 Viscosity Ratio κ for Online, Non-Invasive Lubrication Condition Monitoring of Ball Bearings: An Exploratory Study

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Draft submitted for peer review. Do not cite or distribute without permission.

Keywords: acoustic emission; lubrication condition monitoring; viscosity ratio κ ;
ball bearing; signal processing; gradient boosting; sensor fusion

Abstract

Inadequate lubrication is responsible for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains an unsolved engineering problem. Passive sensor-based approaches are most practical for industrial retrofit because they require no bearing modification, but the signal processing challenge of separating lubrication-relevant information from confounded operating-condition effects has not been systematically characterised for simultaneous acoustic emission (AE) and passive ultrasound (UL) sensing. This paper presents an exploratory study in which AE and passive UL signals are simultaneously acquired from a ball bearing during a continuous speed sweep (1–5985 rpm) at temperatures spanning 33–100 °C, covering the boundary, mixed, and full-film lubrication regimes across 8 783 sweep–sensor pairs per channel. The ISO 281 viscosity ratio $\kappa \in [0.003, 1.64]$ (mean 0.51) is derived analytically from bearing geometry, lubricant properties, and operating conditions as a continuous, physics-grounded regression target. For the AE channel, 26 time-domain and frequency-domain features are extracted from the broadband signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), yielding 104 candidate features; 26 features are extracted from the pre-filtered UL channel (0–20 kHz). A two-stage selection retains 32 AE and 6 UL features after a joint Spearman–Pearson correlation threshold filter and a greedy variance-inflation-factor and pairwise-correlation redundancy reduction. Three regression model families are benchmarked across three sensor configurations using repeated nested five-fold cross-validation ($R = 5$ repetitions, 25 outer scores per model); LightGBM on AE features alone achieves the best hold-out performance ($R^2 = 0.951$, RMSE = 0.078 κ -units). Fusing AE and UL features yields only a marginal improvement ($\Delta\text{RMSE} = 0.0011$) that is not confirmed by holdout significance testing (Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI includes zero). Significance of pairwise model differences is assessed by Nadeau–Bengio corrected repeated k -fold t -tests with Holm–Bonferroni correction for within-sensor comparisons, and by Wilcoxon signed-rank, Diebold–Mariano, and bootstrap ΔRMSE confidence intervals for cross-sensor comparisons. The dataset is made publicly available to support standardised comparison.

Contents

1	Introduction	3
2	Background	6
2.1	Lubrication regimes and the viscosity ratio κ	6
2.2	Physical origins of AE and passive ultrasound signals	7
3	Experimental Setup	7
3.1	Test rig and bearing	7
3.2	Sensor configuration and data acquisition	8
3.3	Operating conditions and κ derivation	8
3.4	Dataset availability	9
4	Signal Processing Pipeline	9
4.1	Signal cleaning	10
4.2	Filter bands	10
4.3	Feature extraction	10
5	Feature Analysis	12
5.1	Correlation with κ	12
5.2	Operating-condition confounding	14
5.3	PCA and regime separability	15
6	Regression Modelling and Statistical Evaluation	15
6.1	Experimental design and holdout split	15
6.2	Repeated nested cross-validation	16
6.3	Statistical significance testing	16
6.4	Results	17
7	Discussion	20
7.1	Why AE outperforms the UL channel	20
7.2	Why fusion provides only marginal gain	20
7.3	Interpreting the amplitude-confounding result	20
7.4	Implications for the κ regression framework	21
7.5	Limitations	21
8	Conclusion	22

1 Introduction

Rolling element bearings are among the most prevalent mechanical components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—whether caused by starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately leads to premature bearing failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received substantially less research attention than vibration-based fault detection and remaining useful life estimation, which are now mature technologies deployed widely in industry. The principal reason is that lubrication-related signal changes are subtler and more entangled with operating conditions than the impulse signatures of surface spalling or raceway cracks. Accurate, online LCM would enable condition-based relubrication: applying lubricant at the right time in the right quantity rather than following time-based schedules that systematically produce both under- and over-lubrication [4].

The dominant approach to bearing condition monitoring is vibration analysis, which is highly effective at detecting macroscopic faults such as raceway spalling and rolling element damage through characteristic frequency analysis. However, vibration-based methods are inherently less sensitive to the subtle tribological changes associated with lubrication film formation and breakdown: under varying speed and load, operating-condition effects dominate the spectral signature and mask the comparatively weak lubrication-dependent component [5, 6]. Temperature monitoring and oil analysis provide complementary information but are limited by indirect measurement, sampling delay, or the need for access to the lubricant reservoir. None of these approaches yields a direct, continuous, and non-invasive indicator of the instantaneous lubrication regime under operational conditions.

Passive, non-invasive sensing methods—acoustic emission (AE) and passive ultrasound (US)—are practically attractive for industrial LCM because they require no bearing modification and can be retrofitted to bearings in service. AE is sensitive to the broadband stress waves emitted by asperity contact events and microcrack propagation across a wide frequency range (typically 50 kHz–4 MHz). Passive ultrasound in the 38–40 kHz band is already industrially deployed in commercial instruments (UE Systems, Sonotec) for threshold-based lubrication assessment: the broadband ultrasonic level is compared against a reference baseline and relubrication is triggered when the level rises by a specified number of decibels [1, 7, 8]. Despite both sensing approaches being in practical use, their signal-processing behaviour has not been characterised simultaneously in a unified regression framework with a physics-grounded lubrication reference variable.

Reference
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Reference?
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Reference

Reference

The sensitivity of AE to lubrication regime has been established across several bearing configurations. Miettinen et al. [9] showed that AE pulse count and RMS distinguish boundary, mixed, and full-film lubrication regimes and are sensitive to base oil viscosity and thickener content in grease-lubricated bearings. Yoshioka and Shimizu [10] demonstrated a compound AE and vibration monitoring system on a grease-lubricated deep-groove ball bearing, using envelope demodulation and statistical AE parameters to characterise lubrication state. Tandon et al. [11] found that AE and shock-pulse sensors outperform accelerometers for detecting grease contamination in a ball bearing—the sensing scenario most closely related to the present study. More recently, Cornel et al. [6] showed that AE detects lubrication-sensitive surface damage such as micro-pitting and smearing up to eight hours earlier than vibration monitoring, and Hou and Zhang [12] derived amplitude-centre and energy-centre features that classify lubrication status across speeds and viscosity levels on a thrust rolling bearing, without a continuous physics-grounded regression target. On signal processing, Renhart et al. [13] proposed octave filter-bank decomposition with RMS–friction product features for AE-based tribological event characterisation in bearing contacts—applying frequency sub-band analysis to contact-type identification rather than to continuous lubrication adequacy regression—and Marticorena and García Peyrano [14] showed that spectral correlation of AE signals resolves lubrication regime changes on a ball bearing under varying load, demonstrating that kinematic and acoustic features together can partially decouple operating-condition confounds. The physical link between lubrication film adequacy and the resulting AE signature has been formalised by Liu et al. [15] and Zhang et al. [16], who developed coupled vibration–AE models incorporating EHL film stiffness and lubricant traction dynamics, establishing that emission intensity and frequency content are governed by the film thickness and the rate of asperity contact. For the passive ultrasound channel, Jakobsen et al. [1, 8] demonstrated that broadband acoustic features from a MEMS microphone sensor carry measurable information about the ISO 281 viscosity ratio κ and established the κ -regression framework that this paper extends. Despite this body of work, no study has simultaneously acquired calibrated AE and passive US from a single ball bearing and quantitatively assessed their complementarity for κ regression under a common experimental protocol—the gap this paper addresses.

A key limitation of existing data-driven LCM work is the reliance on discrete fault labels or qualitative lubrication states rather than physically interpretable, continuous targets. The ISO 281 viscosity ratio κ addresses this directly: it encodes the ratio of the actual lubricant kinematic viscosity at operating temperature to the minimum viscosity required for full elastohydrodynamic (EHL) film formation at the current speed and bearing geometry, and thereby maps the instantaneous operating conditions to a standardised lubrication adequacy index. A model trained to predict κ implicitly learns the full multi-dimensional dependence on speed, temperature, and lubricant properties, and its output

is directly interpretable in terms of established lubrication regime boundaries. Jakobsen et al. [1, 8] demonstrate κ regression from vibration and passive US features using LASSO and neural networks, establishing the methodological foundation that this paper builds upon. Despite its relevance as a continuous, physics-grounded target, the use of κ as a regression objective in acoustic condition monitoring remains largely unexplored.

The complementarity of AE and passive US for LCM has been argued on physical grounds in reviews [5, 17], but has not been quantified experimentally in a unified framework. Whether jointly exploiting both sensor modalities improves κ prediction over either alone is an open question with direct practical implications: additional sensors impose cost and complexity, and are only justified if they provide non-redundant information.

The present study is conducted under controlled conditions using a single bearing configuration, lubricant, and loading, with operating variations limited to rotational speed and temperature. This scope is a deliberate choice: controlled conditions allow lubrication-dependent acoustic behaviour to be isolated from confounds arising from bearing-to-bearing variability and degradation progression. Generalisability to other bearing types, lubricants, and operating environments is an acknowledged limitation and a primary direction for future work.

Contributions

This paper makes the following contributions:

1. **Novel dual-channel experiment.** The first study to simultaneously acquire AE and passive US signals from a ball bearing and regress both, individually and jointly, to κ using a common dataset and analytical ground truth.
2. **Sub-band feature extraction for AE.** Features are computed from the broadband AE signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), enabling model-driven identification of the most informative AE frequency range for lubrication state prediction.
3. **Quantified fusion test.** The first quantitative test of AE–US complementarity in a unified experimental framework, including a positive fusion result that directly addresses an open question in the LCM literature.
4. **Rigorous comparative evaluation.** All model comparisons use repeated nested cross-validation with corrected significance tests (Nadeau–Bengio, Holm–Bonferroni, Wilcoxon, Diebold–Mariano, bootstrap Δ RMSE) that account for the correlation structure of cross-validated estimates, converting the fusion and sub-band comparisons into statistically grounded conclusions rather than point estimates susceptible to inflated confidence.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical background of κ and the physical origins of AE and UL signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal processing and feature extraction pipeline. Section 5 presents the feature analysis results. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings. Section 8 concludes.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

Rolling element bearing contacts operate in one of three lubrication regimes depending on whether the lubricant film is sufficient to fully separate the rolling surfaces. In the full-film elastohydrodynamic (EHL) regime the surfaces are completely separated and asperity contact is suppressed; in the mixed regime the film is partially formed and intermittent asperity contact occurs; in the boundary regime the film is inadequate and metal-to-metal contact dominates. The ISO 281 standard characterises these regimes through the viscosity ratio κ , with boundaries at $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1$ (mixed), and $\kappa \geq 1$ (full film).

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu}{\nu_1(n, d_{pw})}, \quad (1)$$

where ν [$\text{mm}^2 \text{s}^{-1}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [18]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000 \, n^{-0.83} d_{pw}^{-0.5} & n < 1000 \text{ rpm}, \\ 4\,500 \, n^{-0.5} d_{pw}^{-0.5} & n \geq 1000 \text{ rpm}. \end{cases} \quad (2)$$

The operating viscosity ν is computed from the ASTM D341 Walther viscosity–temperature equation fitted to the lubricant's 40 °C and 100 °C reference values, with a standard value of $\lambda = 0.7$. Since ν_1 decreases with increasing rotational speed—faster rotation enhances hydrodynamic film entrainment, requiring less viscosity to achieve full EHL— κ captures the combined effect of speed, temperature, and lubricant grade on lubrication adequacy at a given operating point. Importantly, κ assumes nominally intact lubricant: the Walther equation reflects the properties of fresh grease, so κ does not encode viscosity changes arising from lubricant degradation, contamination, or starvation. Extending the regression framework to track such changes would require either continuous viscosity updates or an alternative target variable.

2.2 Physical origins of AE and passive ultrasound signals

Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings. [Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing.](#) [The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime \[6, 9\].](#) Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors typically operate in the 50 kHz–1 MHz range and require sampling rates exceeding 500 kHz to capture the full frequency content.

Passive ultrasound sensing at 38–40 kHz provides a complementary, lower-frequency view of bearing acoustics. At these frequencies the sensor detects structure-borne acoustic waves whose amplitude is modulated by the contact stiffness and viscous damping changes associated with lubricant film formation: as the film degrades and asperity contact increases, the broadband ultrasonic level at the bearing housing rises [7]. Commercial instruments (UE Systems, Sonotec) exploit this by comparing the level against a reference baseline and triggering relubrication when a defined threshold is exceeded, making threshold-based passive UL the only non-invasive LCM method in routine industrial deployment. However, the level is also sensitive to rotational speed and load independently of lubrication condition, which limits the reliability of threshold-based approaches under varying operating conditions.

The two modalities thus respond to the same underlying phenomenon—increasing asperity contact as the lubricant film degrades—but through different physical mechanisms and at different frequency scales: AE captures individual high-frequency transient events (≥ 50 kHz) from discrete micro-fractures and asperity collisions, while passive UL at 38–40 kHz reflects the lower-frequency bulk acoustic envelope comprising structural resonances, friction noise, and the aggregate effect of contact activity. Both channels therefore carry information about κ , and the question of whether they provide complementary or redundant lubrication-relevant information is addressed experimentally in Sections 3–6.

3 Experimental Setup

3.1 Test rig and bearing

The test rig is a custom-built bearing test stand at CeramicSpeed (Holstebro, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter $d_{pw} = 38.0$ mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) is driven by a

variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt) .

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz and its multiples in the low-frequency range of both sensor signals. These are partially filtered out

3.2 Sensor configuration and data acquisition

Two sensors simultaneously acquired signals from the bearing housing through a shared oscilloscope-based data acquisition (DAQ) system:

AE sensor. A wideband piezoelectric AE sensor is bonded to the bearing housing. Sampling rate: 1.6 MHz (nominally; effective bandwidth verified to 800 kHz by the DAQ). Signal recorded to a digital oscilloscope with voltage range adapted to the signal level (500 mV–1 V per division depending on RPM; see Table ??).

Passive UL sensor. A heterodyned passive ultrasound probe is clamped to the bearing housing. The heterodyning process frequency-shifts the raw acoustic signal into the audible range (<10 kHz); the recorded signal therefore reflects low-frequency acoustic emissions rather than true wideband ultrasound. Voltage range: 50 mV per division. Sampling rate: 1.6 MHz (recorded at the same rate as the AE channel; effective content below 10 kHz).

The oscilloscope samples at a shared rate of 1.6 MHz across both channels, ensuring co-temporal acquisition of AE and passive UL signals. Data are acquired in individual time-series sweeps triggered at each operating condition. Each sweep comprises a fixed number of samples from both channels recorded simultaneously.

3.3 Operating conditions and κ derivation

The experiment follows a protocol which varies the operating conditions through rotational speed (rpm) and housing temperature (°C).

The test profile runs a staircase sweep from 100 to 3000 RPM and back down to 100 RPM, in steps of 100 RPM with a 60-second hold at each step. This is repeated for each of the 13 temperature blocks (40–100°C in 5°C increments), with a 2-minute standstill (RPM = 0) at the start of each block for thermal stabilisation.

Table 1 summarises the operating envelope and resulting κ distribution.

What is the amount?

Do we need NLGI consistency grade?

How is temperature controlled and measured?

Check that this is correct

Explain the process and expected outcome.

How many? How long is sampled?

Table 1: Operating envelope and κ distribution for the retained dataset (8 783 sweeps per sensor). Regime boundaries follow ISO 281: $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1.0$ (mixed), $\kappa \geq 1.0$ (full film).

Quantity	Min	Max
Shaft speed [rpm]	1	5 985
Bearing temperature [$^{\circ}\text{C}$]	33	100
Viscosity ratio κ	0.003	1.644
Regime	Sweeps	%
Boundary ($\kappa < 0.5$)	5 281	60%
Mixed ($0.5 \leq \kappa < 1.0$)	2 552	29%
Full film ($\kappa \geq 1.0$)	950	11%
$\bar{\kappa}$	0.508	
σ_{κ}	0.353	

The κ value for each sweep is computed according to Equations (1)–(2) using the per-sweep measured RPM and temperature, with ν evaluated via the ASTM D341 Walther viscosity–temperature equation. No separate reference measurement of film thickness is available; κ is therefore a model-derived quantity subject to the assumptions of the Hamrock–Dowson framework and Walther’s equation.

Note on regime coverage. The boundary lubrication regime ($\kappa < 0.5$) dominates the dataset (60%) because the bearing spends the largest fraction of the speed ramp at lower speeds where the required viscosity ν_1 is high. Full-film conditions ($\kappa \geq 1.0$) are reached only at higher shaft speeds and account for 11% of sweeps. Model performance in the full-film regime is therefore evaluated on a smaller absolute number of sweeps, which is a limitation discussed in Section 7.

3.4 Dataset availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the analysis pipeline scripts) will be made publicly available on Zenodo upon acceptance. The Python analysis pipeline is configured via a single YAML file and is fully reproducible from the raw HDF5 files.

4 Signal Processing Pipeline

In this section the signal processing pipeline applied to each sweep–sensor pair, comprising signal cleaning, and feature extraction.

- Goal of the pipeline
- Elements included in the pipeline

- Practical implementation (Python version, libraries used, etc.)

4.1 Signal cleaning

4.2 Filter bands

4.3 Feature extraction

Twenty-six features are computed from the cleaned voltage signal for each sweep–sensor pair, comprising 12 time-domain and 14 frequency-domain statistics, as listed in Tables 2 and 3.

Table 2: Time-domain features. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation.

Name	Formula	Description
Peak	$(\max x - \min x)/2$	Half the peak-to-peak range.
RMS	$\sqrt{N^{-1} \sum_n x_n^2}$	Root mean square amplitude.
Std	$\sqrt{N^{-1} \sum_n (x_n - \bar{x})^2}$	Population standard deviation.
Variance	$N^{-1} \sum_n (x_n - \bar{x})^2$	Population variance (square of Std).
Skewness	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^3$	Third standardised central moment; amplitude asymmetry.
Kurtosis	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^4$	Fourth standardised central moment; tail heaviness.
Crest factor	$x_{\text{peak}}/x_{\text{RMS}}$	Peak-to-RMS ratio; sensitive to impulsive events.
Shape factor	$x_{\text{RMS}} / (N^{-1} \sum_n x_n)$	Ratio of RMS to mean absolute value.
Impulse factor	$x_{\text{peak}} / (N^{-1} \sum_n x_n)$	Ratio of peak to mean absolute value.
Margin factor	$x_{\text{peak}} / (N^{-1} \sum_n \sqrt{ x_n })^2$	Peak divided by squared mean square-root amplitude.
Hjorth mobility	$[\text{Var}(x') / \text{Var}(x)]^{1/2}$	Square root of first-derivative to signal variance ratio; proportional to mean frequency.
Hjorth complexity	$M(x')/M(x)$	Ratio of first-derivative mobility M to signal mobility; bandwidth proxy.

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the cleaned signal.

AE sub-band decomposition. In addition to broadband features, three frequency sub-bands are carved from the AE channel using zero-phase Butterworth bandpass filters: (i) 20–500 kHz, capturing the transition from audible to ultrasonic contact dynamics; (ii) 500 kHz–1 MHz, the primary stress-wave propagation band; and (iii) 1–2 MHz, covering high-frequency micro-fracture events. The full 26-feature set is extracted independently for each sub-band, yielding $26 \times 4 = 104$ candidate AE features (broadband plus three sub-bands).

Table 3: Frequency-domain features (companion to Table 2). Notation: $S_k = |X_k|$ one-sided FFT magnitude at bin k ; f_k corresponding frequency; K bin count; $\bar{S} = K^{-1} \sum_k S_k$; $\sigma_S = \text{std}(\{S_k\})$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1} \sum_k f_k S_k$; $\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$.

Name	Formula	Description
Dominant frequency	$f_{\arg \max_k S_k}$	Frequency at which spectral power is maximum.
Spectral mean	$\bar{S} = K^{-1} \sum_k S_k$	Arithmetic mean of the FFT magnitude spectrum.
Spectral std	$\sigma_S = \text{std}(\{S_k\})$	Population standard deviation of the FFT magnitude spectrum.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k / (\sigma_S^3 \Sigma S)$	Frequency asymmetry of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k / (\sigma_S^4 \Sigma S)$	Frequency tail heaviness of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral flatness	$\exp(K'^{-1} \sum_{k:S_k > 0} \ln S_k) / \bar{S}$	Geometric-to-arithmetic mean ratio; 1 for white noise, 0 for a pure tone.
Centre frequency	$f_c = \sum_k f_k S_k / \Sigma S$	Magnitude-weighted mean frequency (spectral centroid).
RMS frequency	$(\sum_k f_k^2 S_k / \Sigma S)^{1/2}$	Magnitude-weighted root mean square frequency.
Freq.-weighted std	$\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$	RMS deviation of bin frequencies from centre frequency (unweighted by magnitude).
Peak frequency	$(\sum_k f_k^4 S_k / \sum_k f_k^2 S_k)^{1/2}$	RMS frequency computed with $f_k^2 S_k$ weighting.
Norm. freq. std	σ_f / f_c	Frequency standard deviation normalised by centre frequency.
Frequency skewness	$\sum_k (f_k - f_c)^3 S_k / (K \sigma_f^3)$	Third spectral moment of frequency deviation from centroid, weighted by magnitude.
Frequency kurtosis	$\sum_k (f_k - f_c)^4 S_k / (K \sigma_f^4)$	Fourth spectral moment of frequency deviation from centroid, weighted by magnitude.
Norm. band-width	$\sum_k f_k - f_c ^{1/2} S_k / (K \sigma_f^{1/2})$	Half-order moment of spectral spread, normalised by $\sigma_f^{1/2}$.

UL channel pre-filtering. The UL channel is pre-filtered to 0–20 kHz by a zero-phase low-pass Butterworth filter before broadband feature extraction, consistent with the effective bandwidth of the heterodyned probe. No frequency sub-bands are computed for the UL channel, giving 26 candidate UL features.

5 Feature Analysis

5.1 Correlation with κ

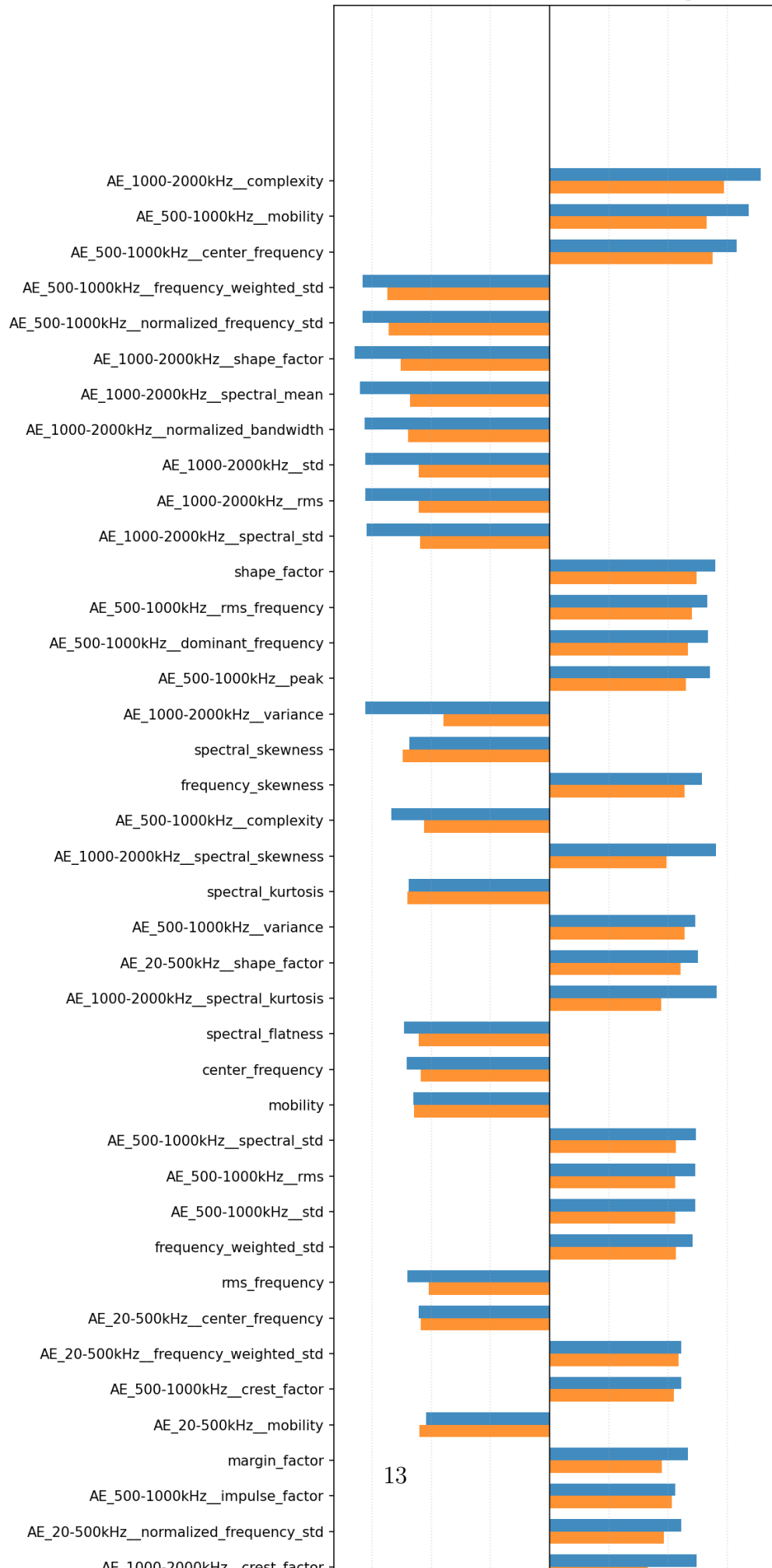
Table 4 lists the top ten features for each sensor ranked by $|\rho(\text{feature}, \kappa)|$ across all 799 sweeps per sensor.

Table 4: Top 10 features by combined Spearman–Pearson rank score with κ , for the AE channel (all sub-bands and broadband combined) and the UL channel. After the Stage 2 redundancy reduction, 32 AE and 6 UL features are retained for modelling; the retained sets are dominated by sub-band features for AE and amplitude/spectral features for UL.

AE (best feature per rank)			UL	
Feature	$ \rho $	$ r $	Feature	$ \rho $
1–2 MHz complexity	0.891	0.736	Spectral mean	0.663
500k–1 MHz mobility	0.841	0.663	Peak	0.629
500k–1 MHz centre freq.	0.789	0.687	RMS	0.607
500k–1 MHz freq.-wt. std	0.789	0.686	Std	0.606
500k–1 MHz norm. freq. std	0.789	0.679	Spectral std	0.607
1–2 MHz shape factor	0.825	0.628	Variance	0.606
1–2 MHz spectral mean	0.801	0.590	Norm. bandwidth	0.607
1–2 MHz norm. bandwidth	0.782	0.599	Freq. skewness	0.539
1–2 MHz std	0.777	0.552	Freq. kurtosis	0.537
1–2 MHz RMS	0.777	0.552	Kurtosis	0.469

The top AE features are drawn overwhelmingly from the 1–2 MHz and 500 kHz–1 MHz sub-bands, with sub-band complexity ($|\rho| = 0.891$) and mobility ($|\rho| = 0.841$) heading the ranking. Broadband AE features rank lower than their sub-band counterparts, indicating that the frequency-specific content of the high-frequency AE signal carries more monotonic information about κ than the full-band aggregate. For the UL channel, amplitude features (spectral mean, peak, RMS, std) dominate at $|\rho| \approx 0.60$ – 0.66 ; the maximum UL correlation ($|\rho| = 0.663$) is substantially lower than the top AE correlation ($|\rho| = 0.891$). Figure 1 and Figure 2 show the full feature correlation rankings for each sensor.

AE — feature correlation ranking



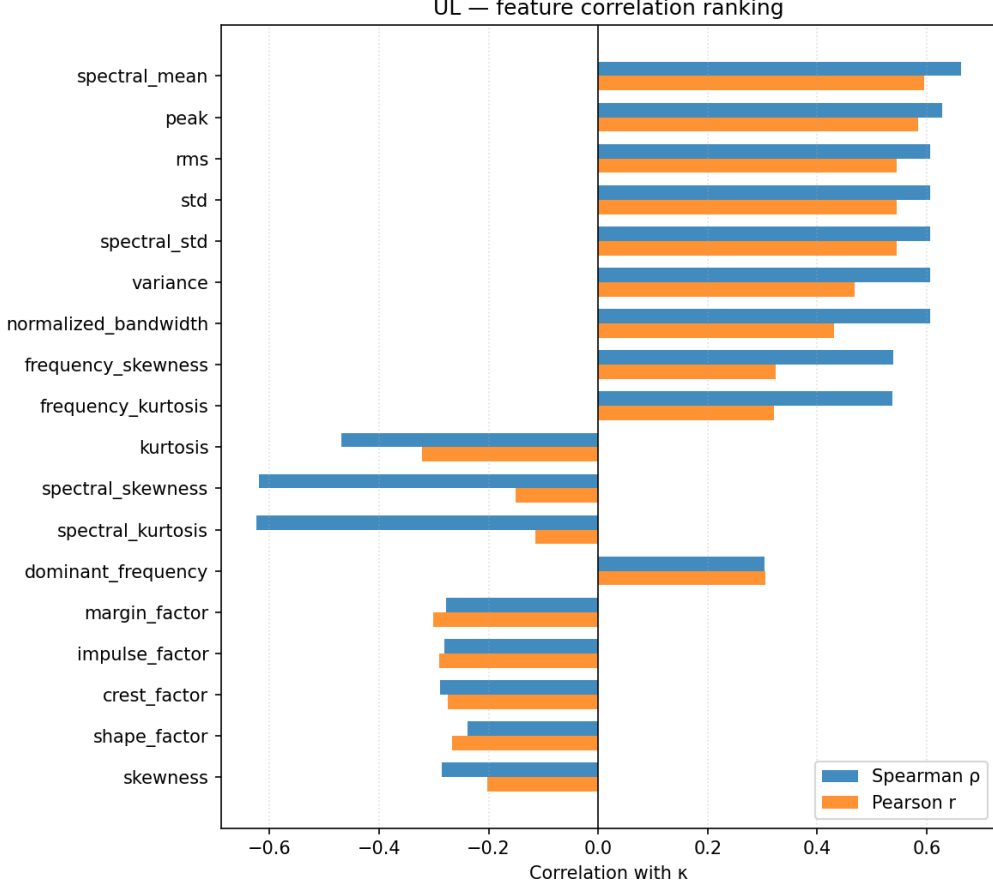


Figure 2: Spearman $|\rho|$ and Pearson $|r|$ between each UL-channel feature and κ , sorted by combined rank. Retained features highlighted.

5.2 Operating-condition confounding

The raw Spearman correlation between broadband signal energy (as a proxy for all amplitude features) and κ is substantially driven by the shared dependence on rotational speed and temperature. Quantitatively, RPM has a Spearman correlation of $\rho = +0.62$ with κ and temperature has $\rho = -0.68$. Higher RPM reduces the required viscosity ν_1 (Eq. (2)) and therefore increases κ , while simultaneously increasing signal energy through more frequent contact events per second. Higher temperature reduces the actual viscosity ν and therefore reduces κ , while also affecting material damping and structural dynamics.

As a consequence, amplitude-dominated features such as RMS, variance, and peak — which rank among the top features by raw Spearman correlation — are partially acting as indirect proxies for rotational speed rather than as direct indicators of lubrication film adequacy. This confounding is an inherent property of any single-operating-point sensor system without an explicit operating condition normalisation step.

Spectral shape features (frequency kurtosis, normalised bandwidth, spectral flatness) are less directly tied to total signal power and are therefore expected to carry a higher

proportion of genuine lubrication-regime information relative to their raw $|\rho|$ rank. The κ regression framework partially addresses this by training models on the full multi-condition dataset, where speed and temperature appear as independent axes of variation and the model can implicitly learn to discriminate amplitude changes due to lubrication from those due to operating conditions.

5.3 PCA and regime separability

Figure 3 shows the first two principal components of the retained AE feature set, coloured by κ regime. Boundary and full-film sweeps are broadly separated in the PC1–PC2 plane, while mixed regime sweeps overlap with both. This confirms that the retained feature set carries regime-relevant discriminative information but that the mixed lubrication regime remains the most challenging region, consistent with the known partial physical overlap between boundary and mixed conditions.

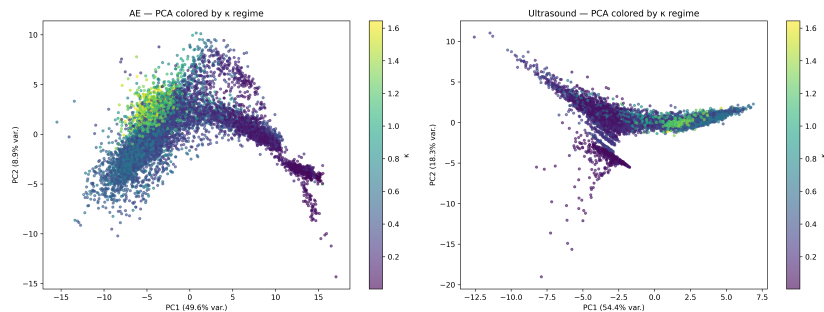


Figure 3: PCA projection of the retained AE feature set, coloured by lubrication regime. Boundary ($\kappa < 0.5$): red; mixed ($0.5 \leq \kappa < 1$): orange; full film ($\kappa \geq 1$): green.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design and holdout split

The dataset is divided into a training set (80%) and a hold-out test set (20%) by sweep-level random sampling, stratified to preserve the κ regime distribution. The hold-out set is withheld entirely from all model development steps and used only for final evaluation.

Three regression model families are evaluated for three sensor configurations (AE only, UL channel only, and combined AE+UL features):

1. **Elastic Net** (regularised linear regression with combined ℓ_1 and ℓ_2 penalty): hyperparameters α and ℓ_1 ratio selected by `ElasticNetCV` with an inner cross-validation loop (9 α values, 6 ℓ_1 ratios).
2. **Degree-2 Polynomial Regression** (polynomial feature expansion followed by Ridge regression via `RidgeCV`): captures pairwise interactions between features while

remaining linear in parameters.

3. **LightGBM** (gradient-boosted decision trees): hyperparameters (learning rate, number of leaves, tree depth, minimum child samples, ℓ_2 regularisation) tuned by Bayesian optimisation using the Optuna TPE sampler (20 trials per inner fold).

6.2 Repeated nested cross-validation

To obtain stable performance estimates that properly account for the variance introduced by the train/validation split, model evaluation uses a repeated nested k -fold cross-validation scheme. The outer loop is repeated $R = 5$ times with independently re-seeded five-fold splits, yielding $R \times k = 5 \times 5 = 25$ outer RMSE scores per model–sensor combination. For LightGBM, each outer fold contains its own inner Optuna optimisation loop, ensuring that the reported outer scores are not contaminated by hyperparameter selection on the outer fold data. For the linear models, hyperparameter selection is handled by `ElasticNetCV` and `RidgeCV` within each outer training partition.

The 25 outer scores form the empirical sampling distribution of RMSE for each model and are used directly in the significance tests described below.

6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels.

Level 1 — Within-sensor comparisons (architecture differences). For pairs of models trained on the same sensor configuration (e.g., Elastic Net vs. Polynomial on AE), the corrected repeated k -fold t -test of Nadeau and Bengio [19] is applied to the 25 paired RMSE scores. This test corrects the standard paired t -test for the positive correlation induced by shared training data across repeated folds. The correction factor is:

$$\hat{\sigma}_{\text{corr}}^2 = \left(\frac{1}{n_{\text{test}}} + \frac{n_{\text{train}}}{n_{\text{test}}} \right) \hat{\sigma}^2, \quad (3)$$

where $\hat{\sigma}^2$ is the sample variance of pairwise RMSE differences and $n_{\text{test}}/n_{\text{train}}$ is the test/train size ratio. Multiple within-sensor comparisons are adjusted by the Holm–Bonferroni step-down procedure at $\alpha = 0.05$.

Level 2 — Cross-sensor comparisons (best per sensor configuration). For the best model from each sensor configuration (selected by lowest mean CV RMSE), pairwise differences on the common hold-out set are assessed by:

- The Wilcoxon signed-rank test on paired residuals (non-parametric);
- The modified Diebold–Mariano test [20] for equal predictive accuracy under squared-error loss;

- A bootstrap Δ RMSE 95% confidence interval (10 000 resamples with replacement).

Cross-sensor p -values are also corrected by Holm–Bonferroni.

6.4 Results

Table 5: Regression model performance. CV RMSE: mean $\pm$ std of outer RMSE across 25 repeated nested CV folds ($n_{\text{train}} = 7\,026$ per fold). HO: hold-out test set ($n = 1\,757$). Best hold-out R^2 in bold. Bayesian Ridge (BR) was trained on the full training set and evaluated on hold-out only (not included in the nested CV framework) and is shown for reference.

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0767 ± 0.0040	0.953	0.050	0.076
LightGBM	AE	0.0780 ± 0.0037	0.951	0.051	0.078
LightGBM	UL	0.2233 ± 0.0060	0.600	0.155	0.222
Elastic Net	Combined	0.1556 ± 0.0051	0.830	0.111	0.145
Elastic Net	AE	0.1568 ± 0.0052	0.826	0.113	0.147
Polynomial	Combined	0.1568 ± 0.0048	0.803	0.113	0.156
Polynomial	AE	0.1646 ± 0.0044	0.775	0.123	0.167
Polynomial	UL	0.2577 ± 0.0056	0.477	0.193	0.254
Elastic Net	UL	0.2783 ± 0.0068	0.396	0.210	0.273
BR (ref.)	Combined	—	0.847	0.103	0.137
BR (ref.)	AE	—	0.830	0.110	0.145
BR (ref.)	UL	—	0.536	0.181	0.239

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations, with AE-only reaching $R^2 = 0.951$, RMSE = 0.078 κ -units. On a target range of [0.003, 1.64] with mean 0.51, this corresponds to a relative RMSE of approximately 15% of the target range. Linear and polynomial models on AE features achieve $R^2 \approx 0.77$ –0.83, confirming a non-linear feature– κ mapping that gradient boosting exploits more effectively.

Sensor comparison. AE features substantially outperform UL-channel features across all model families. The LightGBM AE–UL gap ($\Delta R^2 = 0.35$) is confirmed as highly significant (Nadeau–Bengio $p < 10^{-22}$; Wilcoxon $p < 10^{-166}$; Δ RMSE = 0.144, 95% CI [0.133, 0.157]). This reflects the richer frequency content and more direct physical sensitivity of the AE sensor to asperity contact events.

Sensor fusion yields a marginal, statistically unconfirmed improvement. LightGBM on the combined AE+UL feature set achieves $R^2 = 0.953$, marginally above the AE-only result (0.951). The within-CV corrected t -test rejects equality ($p = 0.024$), but hold-out tests do not confirm the improvement: Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap Δ RMSE = 0.0011 with 95% CI [−0.0009, 0.0033] spanning zero. For the linear models, the combined feature set is similarly on par with AE alone (Elastic Net:

R^2 0.830 vs. 0.826; Polynomial: 0.803 vs. 0.775). The marginal fusion gain is consistent with the UL channel contributing a small amount of non-redundant information at low-frequency.

Model consistency. The close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting. The Bayesian Ridge reference results (hold-out only) align closely with Elastic Net, confirming that the linear model results are not sensitive to the choice of regularisation method.

Figure 4 shows predicted vs. actual κ for the hold-out set for all model families under the AE-only configuration.

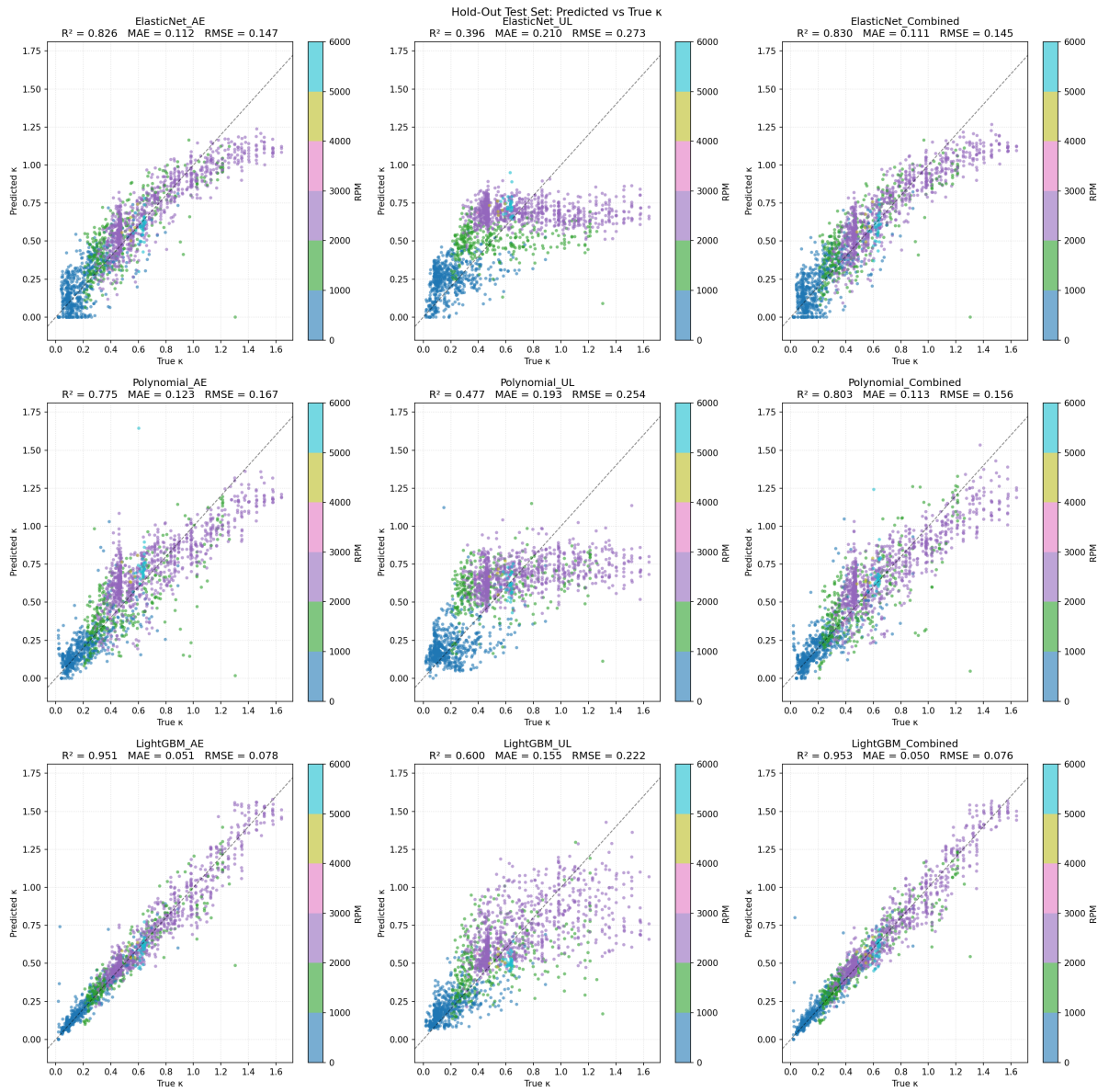


Figure 4: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

SHAP-based feature importance. SHAP (SHapley Additive exPlanations) values computed via TreeExplainer identify which AE sub-band features drive the κ predictions. The dominant contributor by a large margin is *1–2 MHz complexity* (mean $|\text{SHAP}| = 0.162$), followed by *500 kHz–1 MHz centre frequency* (0.051) and *1–2 MHz mobility* (0.038). The 1–2 MHz band complexity feature is also the only feature ranked in the top 10 across all three model types (cross-model agreement, Table 6), making it the most robust single predictor of κ in the dataset. Figure 5 shows the full importance ranking.

Table 6: Cross-model SHAP agreement for AE features: features appearing in the top 10 by mean $|\text{SHAP}|$ for more than one model type.

Feature	ElasticNet	Polynomial	LightGBM	Models in top 10
1–2 MHz complexity	1	1	1	3
1–2 MHz mobility	2	–	3	2
500k–1 MHz spectral std	3	–	10	2
Broadband freq. skewness	10	–	4	2
500k–1 MHz centre freq.	–	10	2	2
500k–1 MHz dominant freq.	–	4	7	2

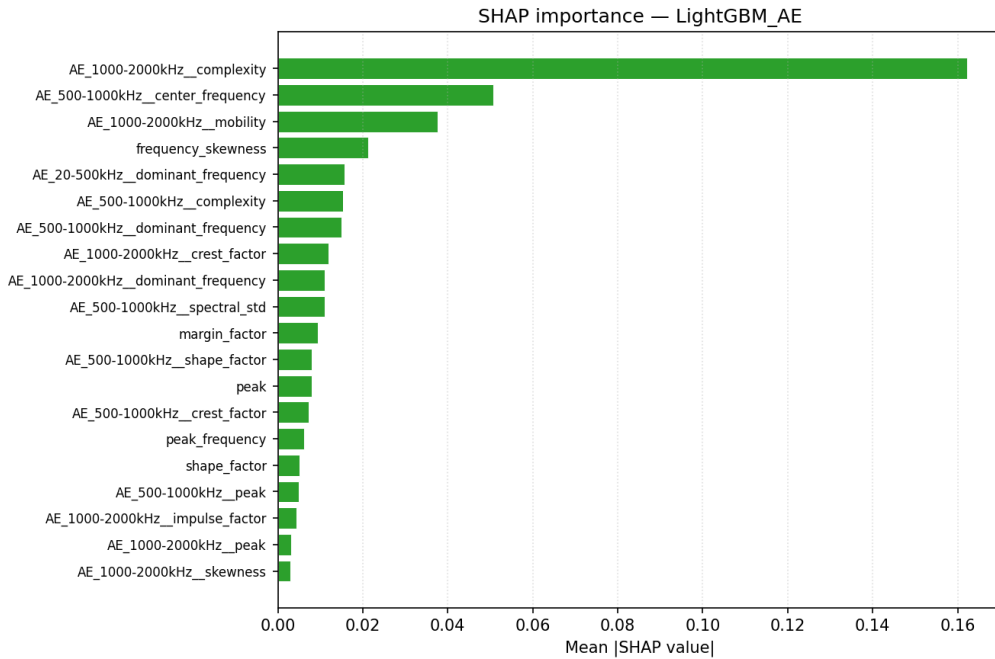


Figure 5: SHAP feature importance (mean $|\text{SHAP value}|$) for the LightGBM model trained on AE features. Features from frequency sub-bands are prefixed with the band label (e.g., AE_20–500kHz_).

7 Discussion

7.1 Why AE outperforms the UL channel

The consistently superior performance of AE over the UL channel across all model families reflects a fundamental hardware difference. The AE sensor captures genuine wide-band stress-wave emissions up to 800 kHz, directly sensitive to individual asperity contact events. The UL channel is heterodyned, limiting its useful content to below approximately 10 kHz, where the signal is a mixture of low-frequency machinery noise, VFD harmonics, and structural vibration. The physical information about microscale contact events is largely absent in this frequency range. Replacing the heterodyned probe with a wide-band passive acoustic or ultrasound sensor would be expected to substantially improve the UL-channel models.

7.2 Why fusion provides only marginal gain

The marginal fusion result ($\Delta R^2 = 0.002$ on the hold-out, not confirmed by Wilcoxon or Diebold–Mariano tests) can be understood from two complementary angles. First, the UL channel retains only 6 features after selection (vs. 32 for AE), and the top UL correlations ($|\rho| \leq 0.66$) are far weaker than the top AE correlations ($|\rho| = 0.89$). The UL channel therefore contributes a small amount of additional information that provides a marginal benefit to the combined model under CV, but the effect is too small to be detectable on the hold-out set (1757 samples). Second, the UL channel’s effective bandwidth (< 10 kHz after heterodyning) means its information content partially overlaps with the low-frequency tail of the AE signal, limiting the independent contribution it can make.

This result is an informative contribution to the LCM literature, where AE–ultrasound complementarity is widely *argued* but rarely *tested*. Whether a wideband passive ultrasound sensor (rather than a heterodyned probe) would provide more independent information – and thus a more substantial fusion gain – remains an open question for future work.

7.3 Interpreting the amplitude-confounding result

The strong raw Spearman correlations of amplitude features (RMS, variance, Hjorth mobility) with κ should be interpreted with caution. As discussed in Section 5.2, rotational speed is the dominant driver of both κ (through ν_1 in Eq. (2)) and broadband signal energy (through the rate of contact events per revolution). A model trained on amplitude features therefore exploits speed as an indirect proxy for κ rather than sensing lubrication-specific signal changes directly.

This confounding does not invalidate the regression results: a model that uses speed-

mediated amplitude changes to predict κ is still predicting κ , and its predictions may be operationally useful. However, it does indicate that the model’s generalisation to operating conditions outside its training distribution (different speed ranges, different bearing loads) may be limited. Future feature engineering should prioritise spectral shape and ratio features (frequency kurtosis, normalised bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information relative to operating condition proxies.

7.4 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM. A hold-out $R^2 = 0.951$ and $\text{RMSE} = 0.078$ κ -units for a single AE sensor represents strong performance across a demanding continuous speed-and-temperature ramp spanning boundary through full-film lubrication conditions. This substantially exceeds the performance reported by Jakobsen et al. [1] in a comparable passive sensing study, which is likely attributable both to the richer sub-band feature set and to the much larger training set (7 026 vs. ~ 640 sweeps) available from the continuous sweep protocol.

The repeated nested cross-validation design provides appropriately conservative performance estimates: the 25 outer RMSE scores capture fold-to-fold variance without the optimistic bias introduced by simple k -fold evaluation with shared hyperparameter search. The corrected significance tests confirm that performance differences between model families are assessed under assumptions that properly account for data re-use, addressing a common methodological weakness in the machine learning for condition monitoring literature.

7.5 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown and is a priority for follow-on experiments.
2. **Heterodyned UL channel.** The passive UL channel is frequency-limited to ≈ 10 kHz. It should not be equated with wideband ultrasound and results should be interpreted in light of this hardware constraint.
3. **Data gap in the mixed lubrication regime.** With only discrete temperature set-points, the mixed regime ($\kappa \approx 0.5$ – 0.7) is underrepresented (24% of sweeps). This is precisely the regime of greatest practical interest for early lubrication deterioration warning.

4. **VFD harmonic contamination.** Switching artefacts at ≈ 5 kHz affect low-frequency spectral features of both channels. Features computed from the UL channel and from the AE band below 10 kHz should be interpreted accordingly.
5. **Model-derived κ reference.** No independent film thickness measurement (e.g., electrical capacitance or active ultrasonic reflectometry) is available. The κ values are model-derived and subject to the Hamrock–Dowson and Walther equation assumptions.
6. **Single session file.** Although 8 783 sweeps per sensor are retained, all data come from a single continuous speed-ramp session of one bearing. Temporal autocorrelation between adjacent sweeps may reduce the effective independence of the training and hold-out sets beyond what the sweep-level split guarantees. Validation across multiple independent sessions and bearings is needed.

8 Conclusion

This paper has presented an exploratory study of simultaneous acoustic emission and passive ultrasound sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **AE sub-band features are strong predictors of κ .** LightGBM on 32 retained AE features achieves hold-out $R^2 = 0.951$, RMSE = 0.078 κ -units. The 1–2 MHz complexity feature is the dominant SHAP contributor and the only feature consistent across all three model types, identifying the high-frequency AE sub-band as the most informative frequency range for lubrication regime prediction.
2. **Sensor fusion provides only a marginal, statistically unconfirmed improvement.** The LightGBM combined model achieves $R^2 = 0.953$ ($\Delta R^2 = +0.002$ over AE alone), but hold-out significance tests (Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI including zero) cannot confirm the improvement. The heterodyned UL channel contributes only 6 retained features with substantially weaker correlations than AE, limiting its independent contribution.
3. **AE strongly and significantly outperforms the UL channel.** The AE–UL gap ($\Delta R^2 = 0.35$ for LightGBM) is confirmed highly significant by all test statistics, reflecting the fundamental bandwidth limitation of the heterodyned UL probe.
4. **Rigorous evaluation via repeated nested cross-validation.** Twenty-five outer RMSE scores per model provide stable, conservative performance estimates; corrected significance tests confirm that LightGBM significantly outperforms linear

models within each sensor configuration, while the Elastic Net–Polynomial difference is not significant for the combined feature set.

5. **The κ regression framework is validated** as a highly effective approach for passive non-invasive LCM, achieving substantially higher performance than prior comparable studies, attributed to the sub-band feature set and the large continuous-sweep dataset.

Future work should address the identified limitations by: acquiring wideband passive ultrasound to replace the heterodyned probe; validating across multiple independent sessions, bearing types, and lubricants; investigating temporal autocorrelation and its effect on hold-out validity; and developing physically motivated sub-band ratio features that are explicitly designed to be robust to operating-condition variation.

CRediT authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the reproducible Python analysis pipeline) will be deposited on Zenodo upon acceptance [DOI to be assigned].

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

Acknowledgements

The author thanks Morten and colleagues at CeramicSpeed (Holstebro, Denmark) for providing the test facility, experimental assistance, and preliminary analysis.

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